

Probability and Statistics



References:

- Richard A. Jhonson, “Probability and Statistics for engineers”
- Jay L. Devore, “Probability and Statistics for Engineering and Sciences”
- Levin, Rubin, “Statistics for Management”
- Tek Bahadur Budhathoki, “Probability and Statistics for engineers”

Chapter 1: Descriptive Statistics and Basic Probability

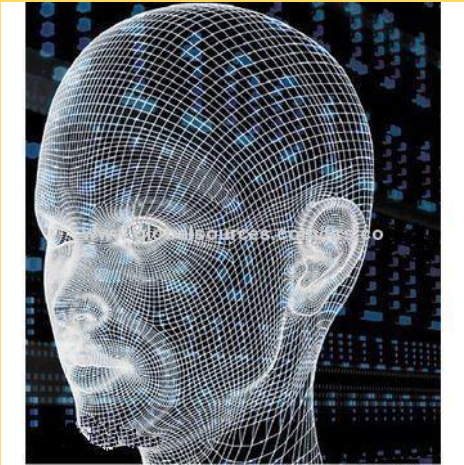
- 1.1 Introduction to statistics and its importance in engineering
- 1.2 Describing data with graphs (bar, pie, line diagram, box plot)
- 1.3 Describing data with numerical measure (measuring center, measuring variability)
- 1.4 Basic probability, additive law, multiplicative law, Baye's theorem

1.1 Introduction/Importance of statistics in engineering field

- The field of statistics deals with the collection, presentation, analysis, and use of data to make decisions, solve problems, and design products and processes.
- It can handle enormous data very effectively, manipulate it for user requirements such that users can draw required conclusions for decision making.
- Because many aspects of engineering practice involve working with data, obviously some knowledge of statistics is important to any engineer.

Statistical techniques can be a powerful tool in :

- designing new products and systems,
- improving existing designs, and
- designing, developing, and improving production processes.
- **Engineers perform tests to learn how things behave under stress, and at what point they might fail.**
- **Engineers apply physical and chemical laws and mathematics to design, develop, improve, test, and supervise various products and services.**



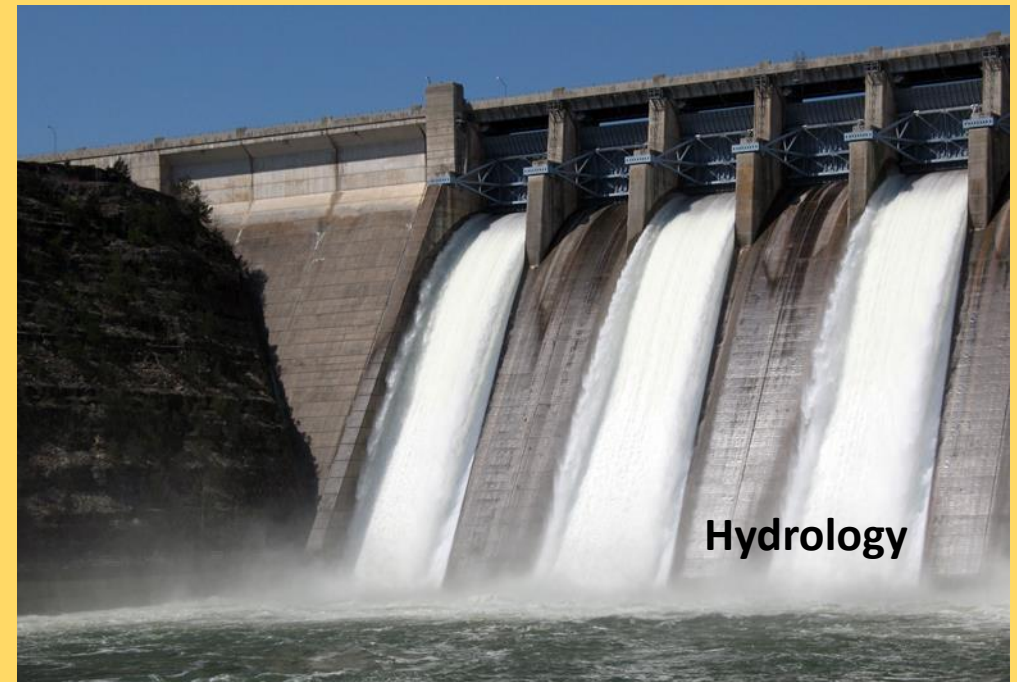
Application of Statistics in Engineering field

- Artificial intelligence
- Data mining (extracting and discovering patterns in large data sets, finding anomalies (irregularities), correlations within large data sets to predict outcomes)
- Automated face recognition,
- automated speech recognition
- automatic language translation
- vision and image analysis
- Automated driving
- strategy games such as chess
- Solar power, wind power are highly probabilistic, and require statistical characterization
- Planning for equipment lifetimes, reliability, probability of outage (power failure), statistics of overload, all require statistical techniques.
- Reasoning for increasing electricity tariff needs good statistical presentation.

Importance of Probability and Statistics in Civil Engineering field

- Civil engineer, in construction of roads, buildings, airports, tunnels, dams, bridges, water supply, and sewage system must consider many factors in the design process like construction cost, expected life time of project, potential environment hazards like earthquakes. Works like supervisory, administrative, research involve lots of uncertainty. To deal with those uncertainties engineers use tools of statistics
- Some major fields in civil engineering are transportation, structural, hydrology, surveying, geotechnical engineering etc.
- **Traffic/transportation Engineering:**
 - to find the hourly volume to be used for design. To fit the distribution of inter-arrival times, ***(The arrival rate is the number of arrivals per unit of time. The inter arrival time is the time between each arrival into the system and the next)***
 - to find the probability that a vehicle stopped at signal will have to wait more than a specified period (traffic engineers perform this kind of analysis to determine where to place traffic signals)
- **Structural engineering:** to find the probability that a design load exceeds certain value, life of structure
- **Surveying and Mapping:** From series of measurement from field, measure “precision” and “accuracy” of instruments used.
- **Hydrology:** to predict stream flow (ft^3/sec) that corresponds to a particular recurrence interval which can be used to design dams (***recurrence interval is an estimate of average time between events, such as an earthquake, flood, landslide, or a river discharge flow to occur***)
- Test of results/findings **statistically** in the research report.

Uses of Probability and Statistics in Civil Engineering field



Statistics is a proven tool for decision making encompassing a wide range of sectors

- engineering,
- management,
- finance,
- scientific research,
- banking,
- economics,
- astronomy,
- meteorology,
- industries and businesses

Engineering also finds widespread applications of statistics in different domains such as

- error analysis,
- production planning,
- quality control and management,
- process control,
- robustness analysis,
- risk assessment etc.

Basic statistics such as **mean, median, mode, standard deviation, regression** as well as advanced analysis tools such as, **z-test, t-test, variance analysis, chi-square test** etc are being widely used across different domains to find first hand solutions to **problems, decision making, optimization, planning and scheduling and data analysis**

1.2 Measure of Central tendency and measure of variation

- Extracting several summarizing numbers which characterize the data set is Average
- Some of the measures of central values are:
 - Arithmetic mean (or mean)
 - Median (average of two middle observation)
 - Mode
 - Quartiles (3 points which divide the whole distribution in 4 equal parts)
 - Percentiles (99 points which divide the whole distribution in 100 equal parts)

Measures of central tendency

- Arithmetic mean ($\bar{X}$) = $\frac{\sum x}{n}$
- Median (Md) = position of $(n/2)$ th observation, after arranging data
- Mode (Mo) = Most repeated observation
- Quartile (Qi) = position of $i(n+1)/4^{\text{th}}$ observation, after arranging data
- Percentile (Pi) = position of $i(n+1)/100^{\text{th}}$ observation, after arranging data

$$\bar{X} = \frac{\sum fx}{N}$$

x = Midpoint

f = frequency

N = Total Frequency

$$M_m = l + \left(\frac{\frac{n}{2} - cf}{f} \right) h$$

Where

l = lower limit of median class,

n = number of observations,

cf = cumulative frequency of class preceding the median class,

f = frequency of median class,

h = class size (assuming class size to be equal)

$$Q_1 = L + \frac{h}{f} \left(\frac{N}{4} - CF \right)$$

$$Q_2 = L + \frac{h}{f} \left(\frac{2N}{4} - CF \right)$$

$$Q_3 = L + \frac{h}{f} \left(\frac{3N}{4} - CF \right)$$

$$M_o = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) h$$

Where

l = lower limit of the modal class,

h = size of the class interval (assuming all class sizes to be equal),

f₁ = frequency of the modal class,

f₀ = frequency of the class preceding the modal class,

f₂ = frequency of the class succeeding the modal class.

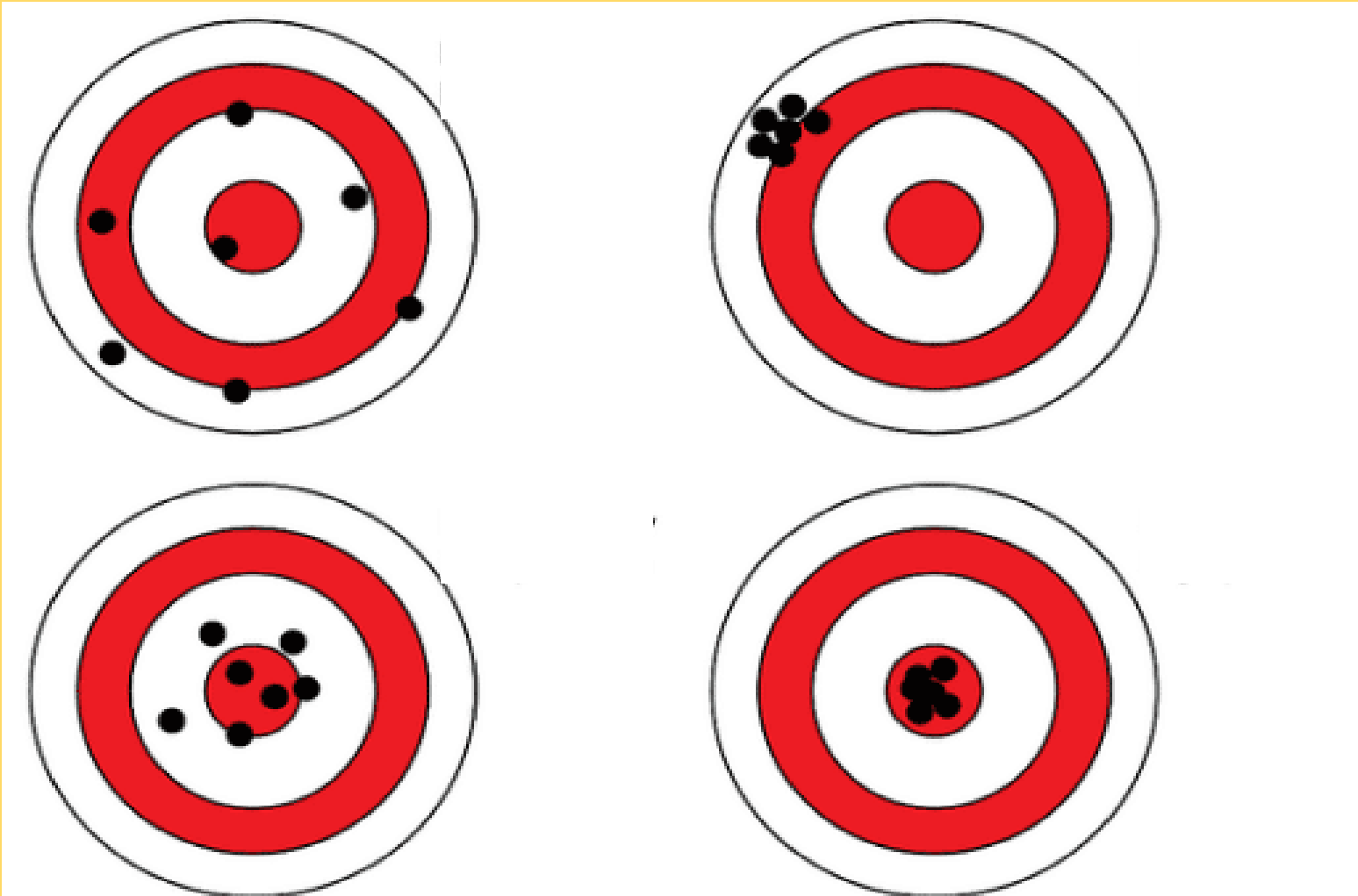
Exercise

A civil engineer monitors water quality by measuring the amount of suspended solids in a sample of river water. Over 11 weekdays, he observed 14, 12, 21, 28, 30, 63, 29, 65, 55, 19, 20 suspended solids (parts per million). Find the third quartile and interpret its meaning.

The following are the temperature of effluent (discharge from a sewage treatment facility) on 24 consecutive days. Draw a box plot and interpret the result.

43, 47, 51, 48, 52, 50, 46, 49, 45, 52, 46, 51, 44, 49, 46, 51, 49, 45, 44, 50, 48, 50, 49, 50

Measures of Variability



- 1) 7, 8, 9, 10, 11
- 2) 3, 6, 9, 12, 15
- 3) 1, 5, 9, 13, 17

Objectives of measure of dispersion

- To measure the reliability of measure of central value. If data are widely dispersed, the central value such as mean is less representative
- To compare the consistency of two or more series of data with respect to variability

Measures of variability

- i) Range**
- ii) Mean deviation**
- iii) Quartile deviation**
- iv) Variance**
- v) Standard Deviation**
- vi) Co-efficient of Variation**

Range = L – S; Large Value – Small Value

Mean Deviation

For Ungrouped Data

$$M.D \text{ (mean)} = \frac{\sum |x - \mu|}{N}$$

For Grouped Data

$$M.D \text{ (mean)} = \frac{\sum f |x - \mu|}{N}$$

$$\text{Quartile Deviation} = \frac{\text{Third Quartile} - \text{First Quartile}}{2}$$

$$\text{Quartile Deviation (Q.D)} = \frac{Q_3 - Q_1}{2}$$

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

σ^2 = population variance

x_i = value of i^{th} element

μ = population mean

N = population size

Population Standard Deviation

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

$$\text{Coefficient of Variation} = \frac{\sigma}{\mu} \times 100\%$$

Standard deviation is the positive square root of average of the square of deviations taken from mean

Population Standard Deviation

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

Variance is the sum of squared distances between the mean and each observation divided by total number of observations.

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

σ^2 = population variance

x_i = value of i^{th} element

μ = population mean

N = population size

Another form of above formula is:

$$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$$

Co-efficient of Variation (C.V)

It is used for comparing the variability of two or more than two series. We calculate C.V. for each series. The series having greater C.V. is said to be more variable than other.

$$\text{Coefficient of Variation} = \frac{\sigma}{\mu} \times 100\%$$

If $x_1, x_2, \dots, x_n$ be n sample observations and C be any non-zero constant,

a) If $y_1 = x_1 + C, y_2 = x_2 + C, \dots, y_n = x_n + C$, then $s_y^2 = s_x^2$

b) If $y_1 = Cx_1, y_2 = Cx_2, \dots, y_n = Cx_n$, then $s_y^2 = C^2 s_x^2$ i.e. $s_y = |C|s_x$

Suppose five weeks of average prices for Stock A are 57, 68, 64, 71 and 62. The mean and standard deviation is 64.40 and 4.84. The coefficient of variation for Stock A is

$$CV_A = \frac{\sigma_A}{\mu_A}(100) = \frac{4.84}{64.40}(100) = .075 = 7.5\%$$

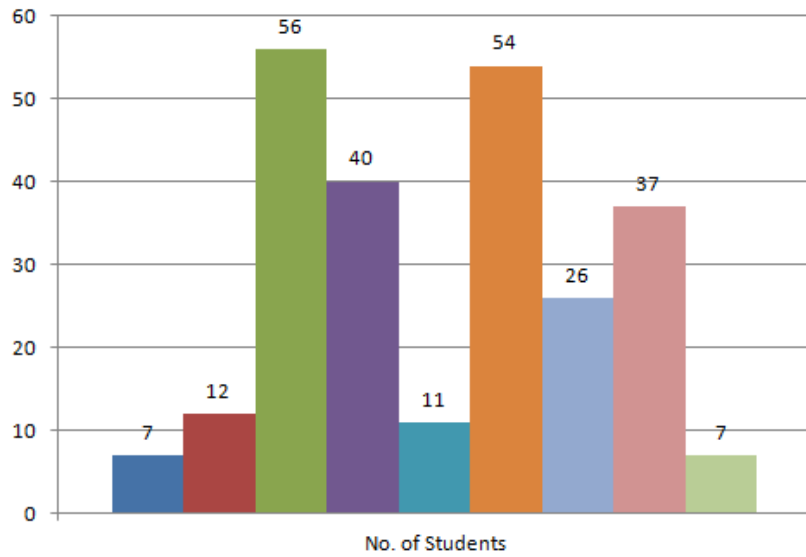
The average price for Stock B five weeks are 12, 17, 8, 15, and 13. The mean for stock B is 13.00 with a standard deviation of 3.03. The coefficient of variation can be computed for stock B as:

$$CV_B = \frac{\sigma_B}{\mu_B}(100) = \frac{3.03}{13}(100) = .233 = 23.3\%$$

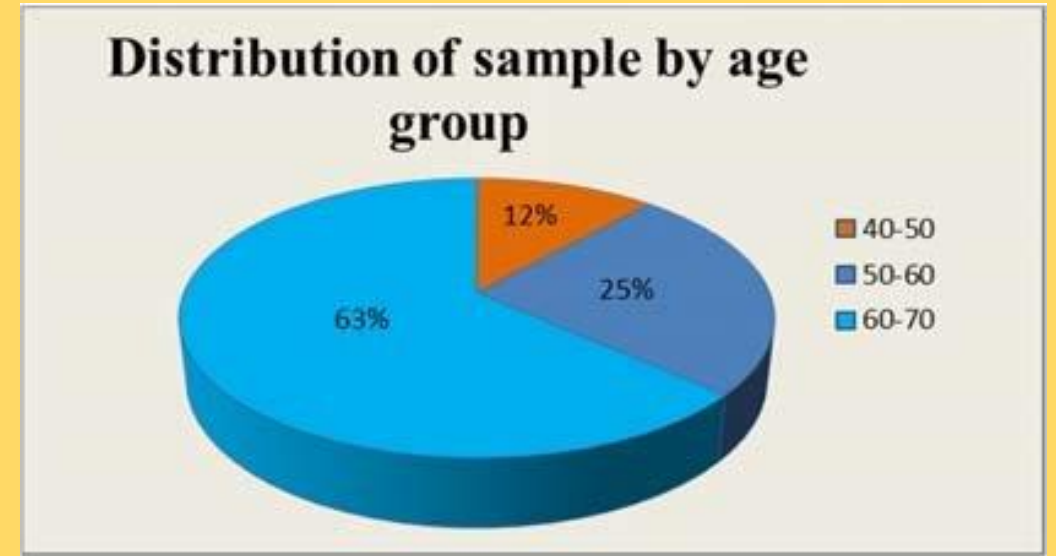
With the standard deviation as the measure of risk, stock A is more risky over this period of time because it has a larger standard deviation. However, the average price of stock A is almost five times as much as that of stock B. Relative to the amount invested in stock A, the standard deviation of 4.84 may not represent as much risk as the standard deviation of 3.03 for stock B, which has an average price of only 13.00.

The coefficient of variation reveals the risk of a stock in terms of the size of standard deviation relative to the size of the mean (in percentage). Stock B has a coefficient of variation that is nearly three times as much as the coefficient of variation for stock A. Using coefficient of variation as a measure of risk indicates that stock B is riskier.

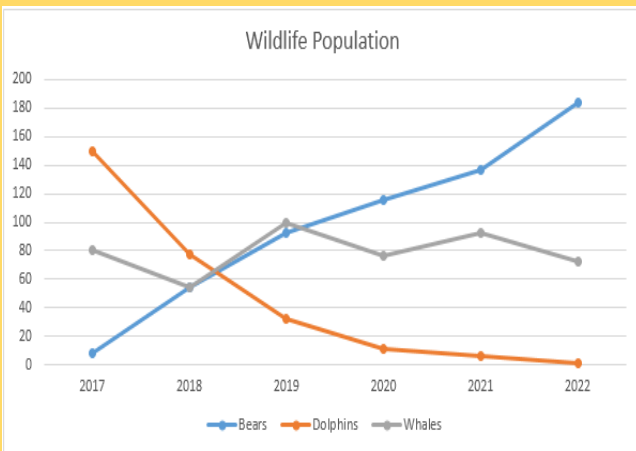
1.3 Graphical Representation of Data:



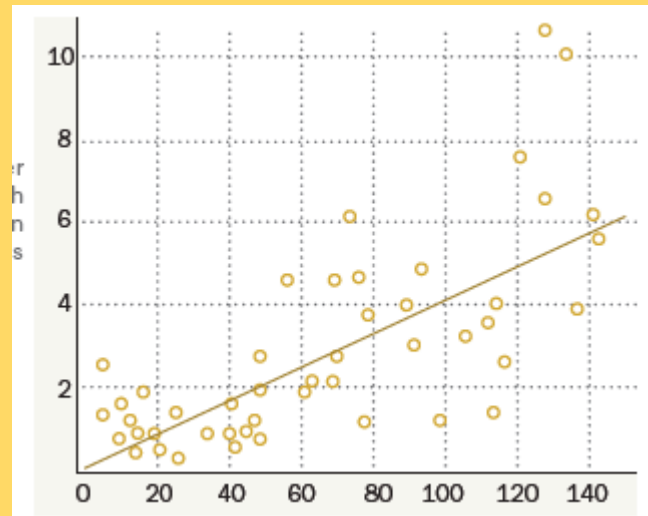
Bar diagram/Histograms



Pie diagram



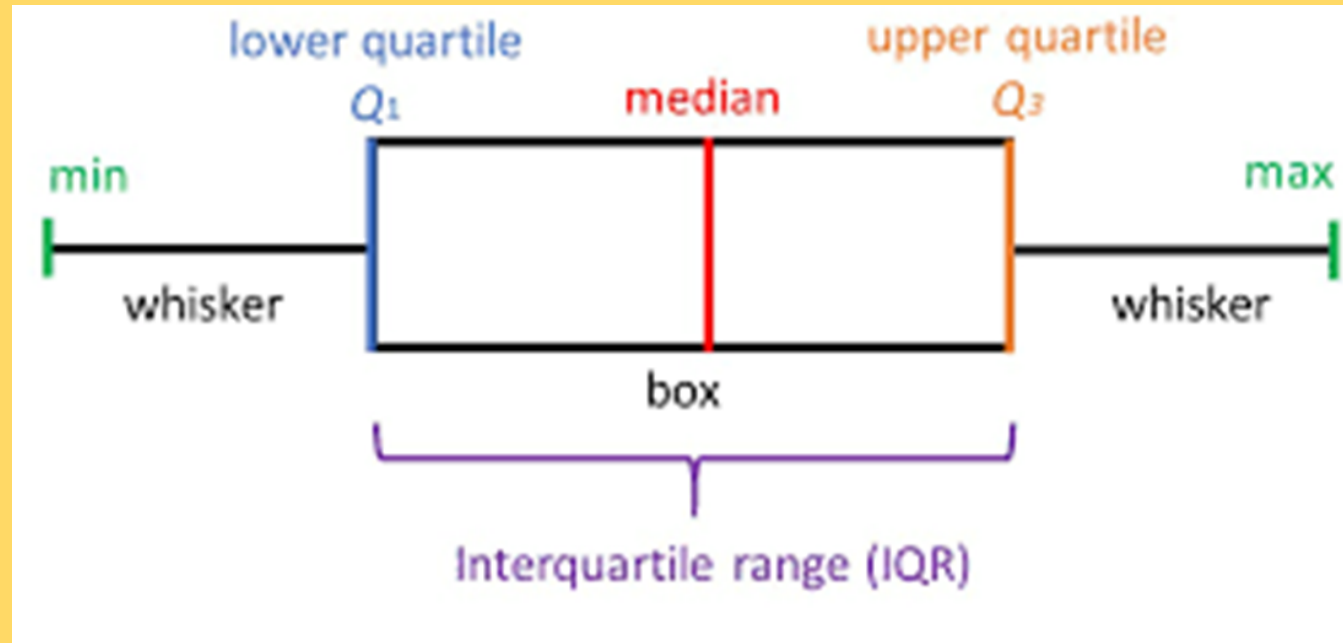
Line diagram



Scatter Plot

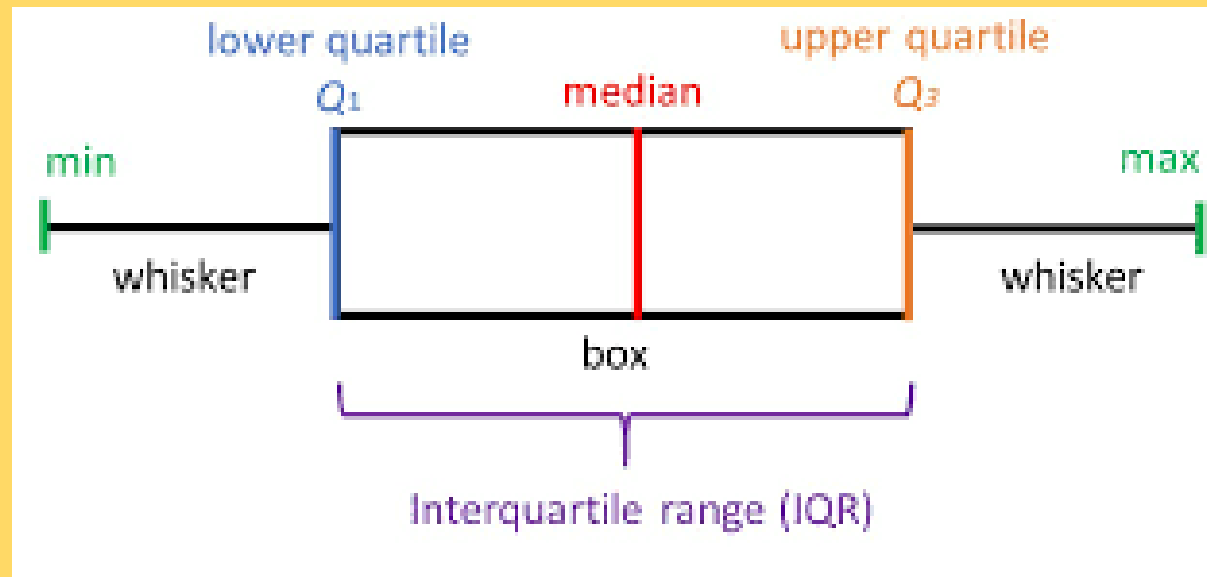
Box plot:

- displays the three quartiles, the minimum, and the maximum of the data on a rectangular box
- aligned either horizontally or vertically
- encloses the interquartile range (IQR) ($Q_3 - Q_1$) with the left or lower edge at the first quartile, and the right or upper edge at the third quartile
- the second quartile is a line drawn through the box
- A line, or whisker, extends from each end of the box



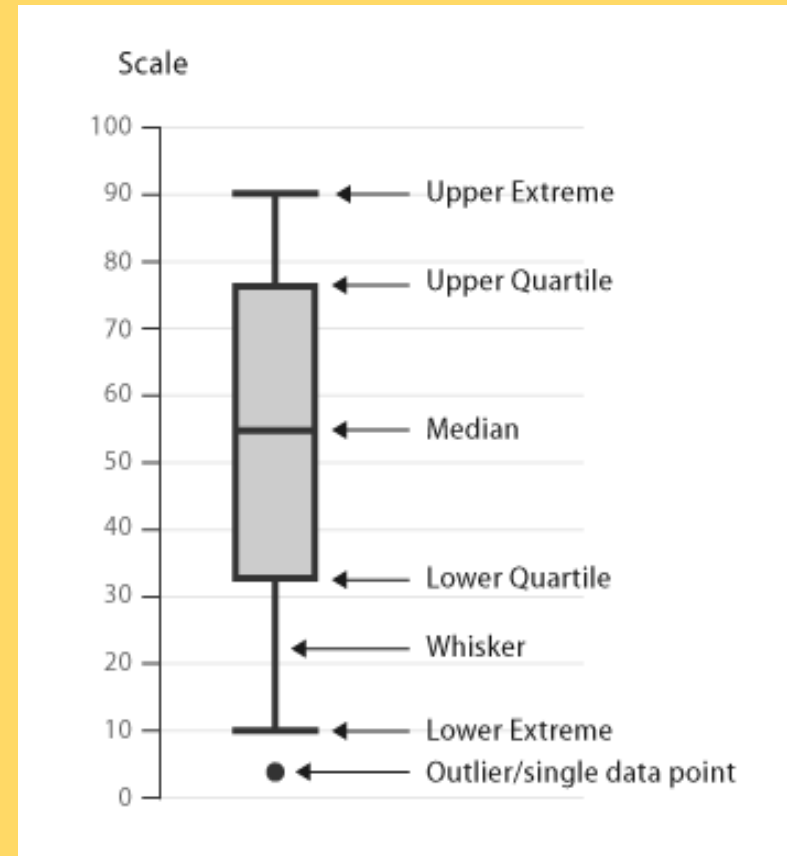
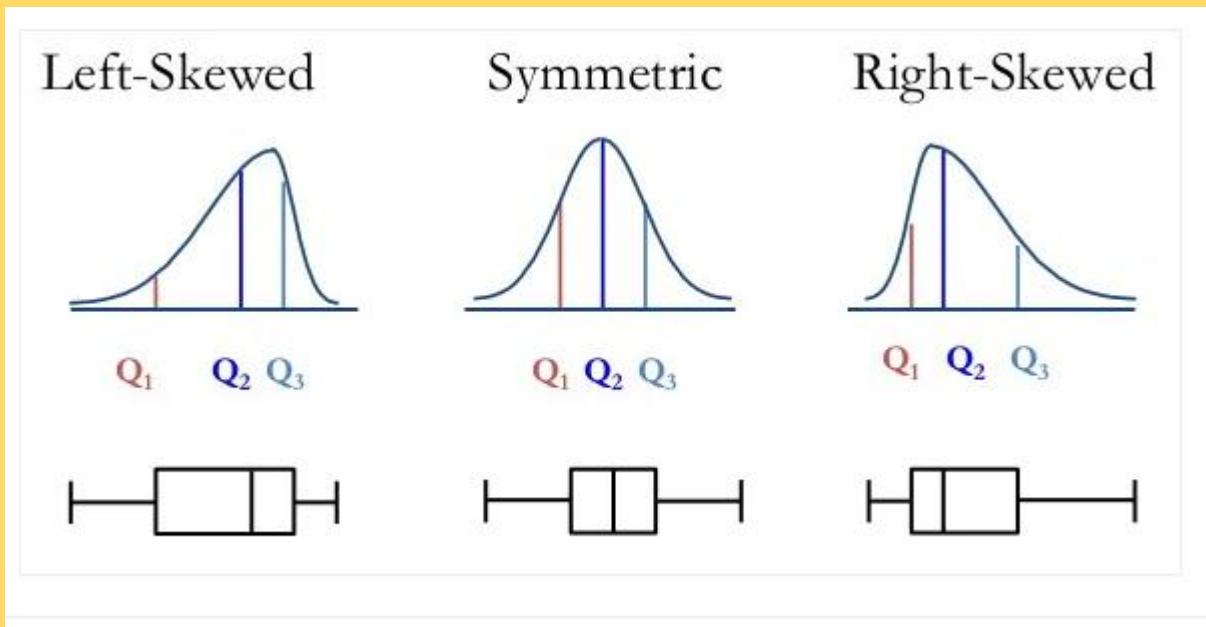
Box plot:

- The lower whisker is a line from the first quartile to the smallest data point.
- The upper whisker is a line from the third quartile to the largest data point.
- Values higher than $Q3 + 1.5 \times IQR$ or lower than $Q1 - 1.5 \times IQR$ are considered outliers and are plotted above the top whisker or below the bottom whisker.



Box plot describes following features of data:

- Center
- Spread
- the extent and nature of any departure from symmetry,
- identification of “outliers,” observations that lie unusually far from the main body of the data



1.4 Basic Probability Concepts

- Probability is the chance something will happen
- expressed as fractions or as decimals between 0 and 1
- Probability zero means something can never happen
- Probability 1 indicates that something will always happen.

Basic Terminologies

- **Trial and event**

- An experiment which is performed under identical conditions gives several possible outcomes.
- The experiment is known as Trial and outcomes are known as events.
- Example, tossing a coin is trial and getting head or tail is an event.

- **Exhaustive event**

- also called sample space
- Total number of possible outcomes in any trial.
- Example, in tossing a coin, there are two exhaustive cases.
- In drawing two cards from a pack of cards, exhaustive cases are ${}_{52}C_2$.

- **Favorable event**

- It is the number of outcomes which requires the happening of the event.
- Example, in throwing two dice, favorable cases of getting sum 5 are (1,4), (4,1), (2,3) & (3,2) i.e. 4 cases.
- In drawing a card from a pack, favorable cases for drawing red card is 26, drawing spade is 13 etc.

- **Mutually exclusive events**

- In any trial, if two or more events cannot happen simultaneously, they are called mutually exclusive events.
- Example: coin toss, getting even and odd numbers in dice

• Equally likely events

- there are equal chances of happening the events
- In tossing a coin (unbiased), there is equal chance of getting head or tail

• Independent events

- the happening of one event does not affect the happening of other
- getting a head in first toss is independent of getting head in second toss.
- if we draw a card from a pack and replace it before second draw, then result of second draw is independent of first draw

- If a trial have 'n' exhaustive cases and 'm' of them are favorable to the happening of an event E, then probability of happening event E is given by,

$$P(E) = \frac{\text{favorable number of cases}}{\text{Exhaustive number of cases}} = \frac{m}{n}$$

If $P(E) = p$, then probability of not happening event E is $1-p = q$

➤ $0 \leq p \leq 1$

➤ $p + q = 1$

➤ $P(E) = 1$, E is called sure event

➤ $P(E) = 0$, E is called impossible event

- **Relative Frequency Approximation of Probability**

Conduct (or observe) a procedure a large number of times, and count the number of times that event A actually occurs. Based on these actual results, $P(A)$ is estimated as follows:

$$P(A) = \frac{\text{Number of times A occurred}}{\text{Number of times trial repeated}}$$

- **Basic Rule of counting**

If there are m ways of arranging one thing and n ways of arranging other thing, then there are $m \times n$ ways of arranging both actions.

- **Permutation**– Ordered sequence of r objects taken from n distinct objects.

$$P(n,r) = \frac{n!}{(n-r)!}$$

- **Combination**– Any unordered subset of size r taken from n distinct objects.

$$C(n,r) = \frac{n!}{(n-r)! r!}$$

Properties of probability

- 1) For any event A , $P(A) = 1 - P(\bar{A})$, where $\bar{A}$ is complementary event of A .
- 2) If A and B are mutually exclusive, then $P(A \cap B) = 0$.
- 3) Law of addition of probabilities: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- for any three events, $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$
- 4) Law of multiplication

Law of Multiplication

- Law of multiplication

Case I. When the events are dependent (Conditional Probability)

The probability of happening an event A, when it is known that some other event B has occurred is called conditional probability. It is denoted by $P(A/B)$ and given by

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$\text{similarly, } P(B/A) = \frac{P(A \cap B)}{P(A)}$$

Case II. When the events are independent

If the happening of one event does not affect the happening of other event, then the events are said to be independent of each other. Two events A and B are independent if $P(A/B) = P(A)$ or $P(B/A) = P(B)$

- If A and B are independent events, then
- $P(A \cap B) = P(A/B) * P(B) = P(A) * P(B)$

$$\text{So, } P(A \cap B) = P(A) * P(B)$$

Example

In a small town, population was categorized as follow

	Employed	Unemployed
Male	460	40
Female	140	260
Total	600	300

One person is selected at random.

- a) If the person is found to be employed, what is the probability that the person was male?
- b) If the person was female, what is the probability that she was unemployed?
- c) What is the probability that the person was employed?
- d) What is the probability that the person was male?

EXERCISE

1. A chemical company has records which shows the following analysis of its 200 workers. If a worker is selected at random for the training from the company, find the probability that:
 - a) He/she has the qualification of Masters
 - b) He/she has Bachelors given that he/she is below 25 years of age
 - c) He/she is above 35 years, given that worker has master's qualification

	Age			
Qualification	Below 25	25-35	Above 35	Total
Masters	10	30	10	50
Bachelors	90	20	40	150
Total	100	50	50	

2. In a group of equal number of men and women, 80% men and 95% women are employed. A person is selected at random and found to be employed. What is the probability that he/she is
 - a) Woman?
 - b) Man?

Baye's Theorem

- Let $E_1, E_2, \dots, E_n$ be n mutually exclusive events with $P(E_i) \neq 0, i = 1, 2, \dots, n$. Then for any arbitrary event A which is subset of Sample space S , such that $P(A) > 0$, we have
- $$P(E_i/A) = \frac{P(E_i \cap A)}{P(A)} = \frac{P(A/E_i) * P(E_i)}{P(A)} = \frac{P(A/E_i) * P(E_i)}{\sum P(A/E_i) * P(E_i)}$$
- The probabilities $P(E_1), P(E_2), \dots, P(E_n)$ are called priori probabilities.
- The probabilities $P(A/E_1), P(A/E_2), \dots, P(A/E_n)$ are called likelihood probabilities.
- The probabilities $P(E_1/A), P(E_2/A), \dots, P(E_n/A)$ are called posterior probabilities